

Optimizing Reward Under Uncertainty: Shifts in Reliance on Reward History vs Sensory Evidence

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Introduction (Written by: Claire Min; Edited by: Coleman Klapheke)

A classic experiment used to test learning and decision-making under uncertainty is the bandit task. In the bandit task, participants are presented with a set of options that vary in assigned reward probability, but the participants are not told the probabilities associated with each option. Consequently, participants must adapt based on the outcomes of their choices to maximize how much rewards they receive. They must decide whether to continue choosing a previously selected option (exploit) or sample alternative options (explore). The behavioral readouts of choice behavior in the bandit task provides insight into how uncertainty and reward history shape goal-oriented decision-making^{1, 2, 3}.

Random-dot motion (RDM) is another paradigm commonly used in decision-making experiments. Participants are shown a cluster of moving dots, with a fraction moving in a specific direction (such as left or right) while the remaining dots move randomly. Depending on the paradigm, either a larger (high coherence) or smaller (low coherence) fraction of dots move in the same direction, and participants must correctly identify an assigned direction of coherent motion. This task is widely used to manipulate sensory certainty and observe response times to understand evidence accumulation during choice behavior^{4, 5}.

The present experiment combines the bandit task with an RDM task to create a game that evaluates how uncertainty and reward interact to influence decision-making. We hypothesized that under perceptual certainty (high coherence), participants would rely more on information from dot-motion direction to try to earn points. However, under perceptual uncertainty (low coherence), we hypothesized that participants would rely more on the previously-rewarded box-location (whether the top or bottom circle had previously yielded more rewards), suggesting a reliance on reward history when sensory evidence is weak.

Methods (Written by: Claire Min, Coleman Klapheke; Edited by: Claire Min, Coleman Klapheke)

Experimental Design

Each participant completed a game consisting of 100 trials. For each trial, participants were shown two circles on a screen, one at the top and one at the bottom, containing RDM with white dots on black backgrounds. Coherence was randomized on a trial-to-trial basis between four levels: 5%, 15%, 25%, and 35%. These values were selected because 15% coherence falls within a low-to-moderate range used in motion-coherence perception studies, while 5% provides high levels of uncertainty⁶. 25% and 35% were chosen as high coherence values to maintain 10 percentage-point

spacing between coherence levels and because they were easily perceivable. Each coherence level was applied randomly across trials and was present in 25 trials per experiment, so there were an equal number of trials for every level. The same coherence level was applied to left and right dot motion on every trial.

The bandit-derived portion of the task was implemented so that leftward and rightward dot-motion directions were the arms of the bandit, with one motion direction corresponding to a 70% chance of receiving a reward and the other corresponding to a 30% chance. Participants were informed that one dot-motion direction had a higher probability of providing a reward upon selection, but they were not informed which direction it was. Whether rightward or leftward dot motion yielded the higher reward probability was randomized for each participant.

To motivate participants to remain focused, respond in a timely manner, and complete the experiment, rewards were provided as points in a game-like setup. After every trial, feedback was provided indicating whether the participant earned a point from that trial. There was also a leaderboard visible after trial feedback and upon completion of the experiment so participants could compare their scores to other participants. After clicking the “Start Game” button, participants were prompted with a two-question survey containing drop-down menus with possible answer selections. The first question prompted participants to select an age group of either 21 and younger, 22 - 30, 31 - 50, or 51 and older. The second question prompted participants to select a gender identity of either male, female, non-binary, or other.

Vibe Coding the Experiment. Google Gemini⁷ was initially prompted to create HTML code based on the project description above, which was formatted as a project proposal for PSYCH 505: Learning and Decision Making at the University of Wisconsin–Madison. Gemini was iteratively prompted to adjust visual design and task functionality based on researcher feedback. The HTML code was uploaded to the researchers’ GitHub⁸, and the program was evaluated and refined through observing changes in the website. Creating the application from only the initially prompted experiment description did not produce an adequately functional task. Because the application is online and used across different electronic devices, Gemini was prompted to lock the game window’s pixel dimensions and apply the device’s refresh rate to the dot-motion animation in the HTML code to reduce lag or refresh-rate misalignment.

The dot-motion display required several literature-informed iterations to better reflect the intended coherence levels. Gemini was prompted with open-source random-dot kinematogram (RDK) codes from two other researchers on GitHub^{9, 10}, as well as a theory paper on generating random-dot motion displays¹¹. After evaluating the display and addressing researcher concerns, four changes were implemented. First, dot motion initially displayed in boxes was changed to circles so participants could not use the straight edges of a box to detect when dots terminated (this is where previously-rewarded box-location was derived). Second, dots were updated using a

noise pattern (brownian) designed to reduce the ability to track individual dots⁹. Third, and in the same vein, dots were given a lifespan so they could not be tracked continuously from side to side within the circle¹⁰. Finally, because pure random assignment of leftward and rightward dot motion to upper and lower circle locations led to repeated location-direction pairings across trials, a more systematic shuffling procedure was implemented to reduce bias from repeated dot-motion directions appearing in the same circle locations.

Collecting Data. Gemini was additionally prompted to route data to Google Firebase¹². The following data were collected for each participant: anonymized subject ID, dot-motion direction yielding a higher reward probability, and final score. Additionally, for every trial, the following data were collected: trial number, coherence level, top and bottom circle dot-motion directions, chosen circle, response time (ms), and whether a reward was received. All data were sent to the researchers' Google Firebase Firestore database¹². For completed experiments, where participants completed all 100 trials, data were collected in an assigned collection in the database. If the experiment was not fully completed, participant data were collected in another assigned collection and not used in the analysis for this experiment. Only five incomplete cases were recorded and excluded from analyses.

Statistical Analyses

Python was used to perform behavioral and model-based analyses utilizing participant-level summary measures. ChatGPT¹³ was utilized to generate graphing code. The analysis code is available at <https://github.com/whoiscolek/rdk-bandit-task>. Exploratory tests like one-way ANOVAs for age and paired *t*-tests on males and females were conducted to test whether key decision metrics differed across gender identity and age groups.

Dot-motion coherence values were coded numerically as 5% = 1, 15% = 2, 25% = 3, and 35% = 4. For every participant and trial, top-box choices were coded as 1 and bottom-box choices as -1. Reward direction was coded as 1 when rightward dot-motion corresponded to the higher reward probability and -1 when leftward dot-motion corresponded to the higher reward probability. The optimal choice was identified for each trial by comparing the dot-motion direction in the top and bottom boxes and identifying which aligned with the reward direction.

RDM Task Verification. Response times were analyzed to determine whether motion-coherence manipulations influenced decision difficulty. Trials with response times greater than 7500 ms were excluded because values above this threshold reached up to 30000 ms and were deemed not reflective of actual choice behavior. For each participant, mean response times were calculated for low and high coherence trials and evaluated using a paired-samples *t*-test.

Reinforcement-Learning Model. To estimate the learned value for the top and bottom options for each participant, a Rescorla-Wagner/Q-learning model was fit

separately for each participant^{14, 1}. Initial action values were set to $Q_{top} = .5$ and $Q_{bottom} = .5$ to reflect neutral expected reward values prior to learning. On each trial, the chosen option, either Q_{top} or Q_{bottom} , was updated using the following equation: $Q_{new} = Q_{old} + \alpha(r - Q_{old})$, where r was 1 if a reward (point) was received and 0 if a reward was not received, and α was the participant learning rate. α was found through maximum likelihood estimation¹. Potential learning rates ranged from 0.00 to 0.99 in increments of .01. For each potential learning rate, the model computed action-value differences for each trial using the following equation: $Q_{diff} = Q_{top} - Q_{bottom}$. Values were then converted into a probability of choosing the top-box option using a logistic function equation: $P_{top} = \frac{1}{1 + e^{-Q_{diff}}}$. The learning rate with the highest summed log likelihood of observed choices was retained as each participant's best-fitting learning rate. After estimating learning rates for every participant, action-value differences were calculated using each participant's optimal α values for all trials and used as reward-history predictor values in subsequent analyses.

Previously-Rewarded Box-Location Choice Alignment. Previously-rewarded box-location (reward-history) alignment was calculated by comparing participants' choices on trials with the choice favored by corresponding action-value differences. If action-value differences ($Q_{top} - Q_{bottom}$) were positive, the learned-value model favored the top option, and if negative, it favored the bottom option. Trials were excluded when the action-value difference was exactly zero. Reward-history alignment values were coded as 1 if the participant chose the option favored by learned reward history and 0 otherwise.

Bandit Task Verification. To ensure that the bandit-task component was functioning as intended, participant-level mean reward-history alignment scores were compared against chance performance (0.50) using a one-sample t -test. Cohen's d was calculated as the difference between participants' average reward-history alignment and chance (0.50), divided by the standard deviation of reward-history alignment values.

Dot-Motion Direction Choice Alignment. Stimulus-indicated (dot-motion direction) choice alignment was calculated by evaluating whether participants chose the box with dot-motion direction matching the direction corresponding to the higher reward probability. Values were coded for each trial as 1 when the chosen option's dot-motion direction matched the higher rewarded direction and 0 otherwise. Participant-level means were computed separately for high and low coherence trials. Low-versus-high coherence comparisons were evaluated using paired-samples t -tests to determine whether reliance on perceptual evidence differed across coherence conditions.

Previously-Rewarded Box-Location vs. Dot-Motion Direction Choice Alignments. To examine the relative reliance on learned reward history versus sensory evidence, the difference between reward-history alignment and stimulus-indicated alignment was calculated for each trial using the following equation: reward-history alignment - stimulus-indicated alignment. These values were averaged for low and high

coherence conditions for each participant. The difference between these values was then calculated and tested against zero using a paired-samples *t*-test. For graphing, values were sign-flipped so that positive values represented greater reliance on dot-motion direction relative to previously-rewarded box-location history.

Model-Based Decision Weights for Low and High Coherence. A logistic choice model was fitted to each participant to calculate regression coefficients to evaluate how much choices were influenced by previously-rewarded box-location or dot-motion direction³. For each trial, sensory signals were calculated by converting coherence percentages to decimal values (i.e., 15% as .15) and multiplying coherence values by the correct box-location (1 for upper having the higher-rewarded motion direction, -1 for lower). Decision signals were then calculated for each trial using the following equation: $\beta_{\text{previously-rewarded box-location}}(Q_{\text{top}} - Q_{\text{bottom}}) + \beta_{\text{dot-motion direction}}(\text{sensory signal})$. Using a logistic choice rule³, the model converted the decision signal into a probability of choosing the top box-location using $P_{\text{top}} = \frac{1}{1 + e^{-\text{decision signal}}}$. For every participant, the model searched over potential values of $\beta_{\text{previously-rewarded box-location}}$ and $\beta_{\text{dot-motion direction}}$ from -5.0 to 4.9 in increments of .1. The combination of values yielding the highest log likelihood of observed choice was retained as the best fit for that participant's choices. This was done separately for each participant for low and high coherence levels. Paired-samples *t*-tests were used to compare model weights at low versus high coherence.

Model-Based Decision Weights for All Coherence Conditions. Model weights were also estimated separately for all four coherence levels (5%, 15%, 25%, 35%). For every participant and coherence level, the same logistic model was used to estimate regression coefficients. To evaluate how sensory evidence compared in dominance to previous reward history, the model calculated dot-motion direction weight minus previously-rewarded box-location weight for each participant and coherence level. A linear slope was then fit across the four coherence levels for each participant to test whether sensory-evidence weighting changed with stimulus certainty⁴. Participant-level slopes were tested against zero using one-sample *t*-tests. A positive slope indicated that sensory evidence increased in relative influence over previously-rewarded box-locations as dot-motion coherence increased.

Learning Over Time. To understand how participants' performances changed, stimulus-indicated choice alignment was analyzed across time-points throughout the experiment for low and high coherence trials. Trials were divided into four equal temporal bins corresponding to early (first quarter), early-middle (second quarter), late-middle (third quarter), and late (fourth quarter) trials based on the sequence in which trials of that coherence condition occurred. Average stimulus-indicated choice alignment was calculated for each bin across participants. Learning over time was tested by comparing the first and final bins using paired-samples *t*-tests for low and high coherence trials.

Reporting. All inferential tests were performed at the participant level. Paired-samples *t*-tests were used for within-subject low-versus-high coherence comparisons, and one-sample *t*-tests were used for chance-level validation and tests of linear change across coherence levels. Results were reported as mean $\pm$ SEM, test statistic, *p*-value, and Cohen's *d*. For paired comparisons, Cohen's *d* was calculated as the mean of within-subject difference scores divided by the standard deviation of those difference scores, using conventional within-subject effect-size calculations. Optimal choice is defined as the choice that had the highest probability of providing a reward based on motion direction and where it was located on screen. Reward history and previously-rewarded box location are used interchangeably and "stimulus-indicated" refers to the choice corresponding to the dot-motion direction that yields a higher probability of providing a reward. Two tables are available for all calculated statistics, including those reported in the results and many more, in the supplemental materials section at the end of this paper (Table S1, Table S2).

Results (Written by Coleman Klapheke)

A total of 56 participants completed the RDM-bandit task. Of those 56, 33 identified as Male, 22 identified as Female, and 1 identified as Non-binary. Additionally, 12 were in the 21 and younger age group, 32 were in the 22 to 30 age group, 8 were in the 31 to 50 age group, and 4 were in the 51 and older age group.

Task Validation

On all trials, participants chose the option with dot-motion direction corresponding to the higher reward probability significantly above chance, $M = 0.562 \pm 0.022$, $t(55) = 2.815$, $p = .007$, $d = 0.376$. Additionally, participants chose options aligned with reward history significantly above chance, $M = 0.588 \pm 0.010$, $t(42) = 8.896$, $p < .001$, $d = 1.357$. Participants responded significantly faster in high coherence than low coherence trials. Average response time decreased from 1956.228 ± 165.691 ms in low coherence trials to 1804.565 ± 141.572 ms in high coherence trials, $t(55) = 2.371$, $p = .021$, $d = 0.317$ (Fig. 1).

Sensory Evidence Had Greater Influence at Higher Coherence

On high coherence trials, participants chose the option with dot-motion direction corresponding to the higher reward probability significantly more often than on low coherence trials. Stimulus-indicated choice alignment increased from 0.547 ± 0.024 at low coherence to 0.587 ± 0.028 at high coherence, $t(42) = -2.155$, $p = .037$, $d = -0.329$.

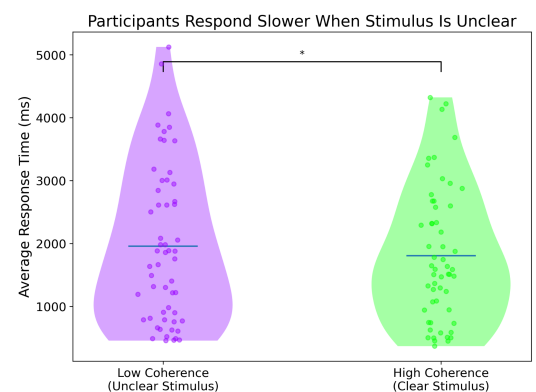


Figure 1. Full spread of participant response times using violin plots and individual jittered points with outliers cut off at 7500 ms. Low coherence (purple) trials had a 200 ms longer response time than high coherence (green) trials.

Learning Occurred Most During Low Coherence

Participants showed the strongest behavioral adaptation when the task was more difficult and perceptual evidence was weaker. During low coherence trials, performance increased from 0.499 ± 0.022 in the first trial bin to 0.571 ± 0.027 in the fourth trial bin, $t(55) = -3.106$, $p = .003$, $d = -0.415$ (Fig. 2). During high coherence trials, performance did not significantly change across the task, increasing from 0.573 ± 0.029 early to 0.604 ± 0.030 late, $t(55) = -1.041$, $p = .302$, $d = -0.139$ (Fig. 2).

Modeling Reveals Coherence-Dependent Shift Toward Sensory Evidence

Across all four coherence levels, the influence of dot-motion direction increased significantly as coherence increased. The slope for dot-motion direction weight minus reward history weight was 4.645 ± 1.716 , $t(55) = 2.707$, $p = .009$, $d = 0.362$ (Fig. 3). As perceptual evidence became more reliable, the regression model estimated greater influence of sensory information over learned reward history.

Reward History Influence Persisted Across Conditions

A higher proportion of choices on low coherence trials aligned with previously-rewarded box-location (0.599 ± 0.013) than on high coherence trials (0.577 ± 0.012), but the difference was not statistically significant, $t(42) = 1.446$, $p = .156$, $d = 0.220$.

Effects of Age and Gender

No significant gender or age differences were observed for choice accuracy (gender: $p = .549$; age: $p = .615$), overall response time (gender: $p = .527$; age: $p = .237$), or the balance between reward-history and stimulus-indicated decision strategies (gender: $p = .179$; age: $p = .505$).

Discussion (Written by: Maxwell Rautmann, Coleman Klapheke; Edited by: Coleman Klapheke)

The present study combined a bandit task with a random-dot motion task to understand how participants incorporate learned reward history when making perceptual decisions at varying levels of certainty. Overall, participants used both

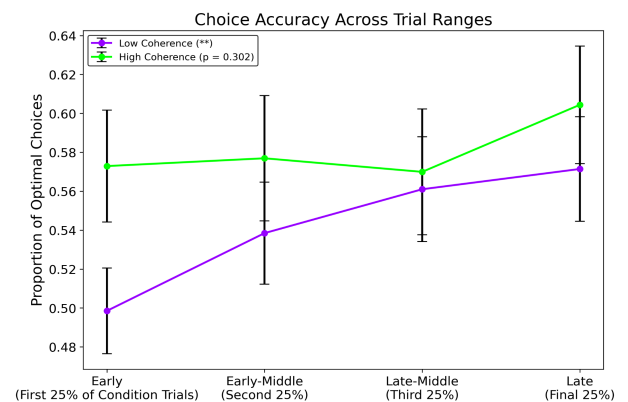


Figure 2. Shows performance across early, middle, and late trials under conditions of low (purple) versus high (green) coherence.

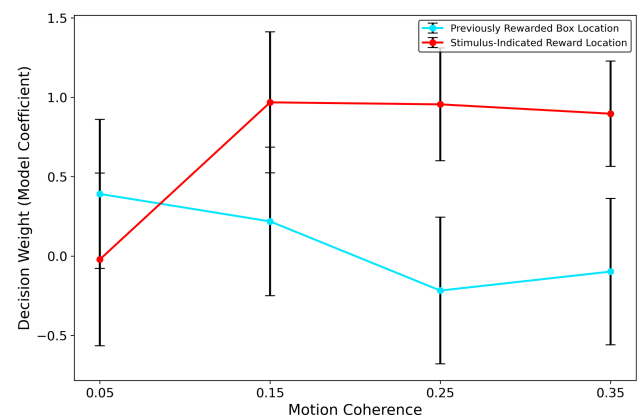


Figure 3. Decision influence across all coherence levels (0.05–0.35) for participants' reliance on Previously Rewarded Box Location (blue line) and Stimulus-Indicated Reward Location (red line). As coherence increased, participants relied more on sensory evidence to guide their decisions, and relied less on reward history.

sources of information to make perceptual decisions. On all trials, participants chose the stimulus-indicated choice and choice that aligned with reward history above chance levels. This suggests that the bandit-derived portion of this task produced reward-guided behavior, similar to previous bandit task studies^{1,2}. Additionally, as coherence levels increased so did participants' reliance on dot-motion direction to guide choices (Fig. 3). As optimal choices were more certain, there was more reliance on current sensory evidence rather than reward-history information. Low coherence trials producing longer response times than high coherence trials validated the random-dot motion task and selected low-high coherence bins (Fig. 1). This is consistent with classic random-dot motion and evidence accumulation studies^{4,5} that show how longer response times reflect uncertainty. This suggests that in the present task, there was greater uncertainty during low coherence trials and clearer evidence at high coherence trials.

Participants chose the optimal choice more on high coherence trials than on low coherence trials (Fig. 2). This finding aligns with previous research that showed that higher motion coherence increases the influence of sensory evidence in decision making^{4,5}. Clear dot motion made it easier for participants to identify the optimal choice and receive a point in the game. Additionally, participants learned how to better make optimal choices over time in low coherence trials but not in high coherence trials (Fig. 2). An explanation for this could be that dot-motion direction at high coherence was so clear that participants began the experiment at their optimal performance rates and thus did not show any improvement on these trials. This finding is consistent with the idea that uncertainty can shift reliance to previous learning to develop a strategy that works when perceptual evidence is unreliable.

The logistic regression model calculations revealed a coherence-dependent shift towards sensory evidence from low to high coherence. Across the four coherence levels, the influence of dot motion increased while the influence of reward-history decreased (Fig. 3). While dot motion had a higher maximum influence overall, at 5% coherence, dot-motion direction had minimal influence relative to higher coherence levels. These findings show that participants were using different strategies at different coherence levels and giving different weights to information based on the reliability of sensory evidence. This supports the idea that decision systems weight information sources based on reliability which is consistent with previous research^{4,5}. Choices aligned more with reward-history at low coherence than at high coherence, but the effect was non-significant. This suggests that participants did not stop using reward history information to guide decisions, but used it less when perceptual decisions were certain and reliance on dot-motion direction yielded rewards.

Demographic analyses revealed no significant differences between any demographic groups in 3 key metrics: choice accuracy, average response time, and the balance of stimulus-guided and reward history-based strategies. This suggests that the

main results were not detectably driven by age or gender in the present sample and provides support for the validity of the main task regarding demographic confounds. It should be noted that race was not reported and thus cannot be ruled out as a potential demographic confound.

Overall, the findings of this study suggest that participants dynamically used learned reward history and sensory evidence during decision-making to select optimal choices and maximize rewards. These findings confirm the original hypothesis that under perceptual certainty, participants would rely on present stimulus information (dot-motion direction). However, they challenge the idea that at low coherence participants only rely on reward history (previously-rewarded box location). The present findings suggest instead that participants use a mixed strategy where they rely on reward history more and sensory information less under perceptual uncertainty. This supports the idea that under uncertainty, decision making requires flexible weighting of prior knowledge as well as current sensory evidence rather than one fixed strategy.

Limitations and Future Directions

While researchers used many prompts with Gemini⁷ ensuring the precision of random dot kinematograms (RDKs), the task was not validated against laboratory-grade dot-motion software or experimental-grade parameters. The use of four discrete coherence values allowed for efficient data collection and analysis, but was not as informative as a gradient scale would have been. More coherence values could have identified a tipping point where subjects transition from perception-based decisions to reinforcement-based decisions. Because the task had only two spatial locations, chance choice performance was 50% from the start, which may have created a bias where reward-history appeared more influential than it actually was. There were no controls for the environment as this experiment was a web-based application. The lack of environmental controls could have introduced uncontrolled variability for any given environment our experiment was performed in. Since this was done on a multitude of devices, there could have been a number of internet service or device compatibility confounds. There were no pauses or breaks during trials and fatigue-related slowing could have biased response-time measures, especially on low-coherence trials.

A future study could address all of these limitations. Consulting with experts in dot-motion physics and human perception to determine the right calculations for dot motion and variables to test for would yield more accurate results. Using four dot-motion directions and four different probabilities would provide more details into participants' decision-making and clarify the role of previously-rewarded box location by distancing it from being a chance decision. Performing this study in a controlled laboratory environment would allow participants to complete this study on consistent hardware and software, take controlled pauses, and have a designed environment free from distractions.

Potential Neural Correlates. Our behavioral results reflect a coherence-dependent strategy shift that is consistent with the idea that the brain may weight information sources based on relative reliability¹⁵. As motion coherence increased, we observed a progressive transition toward sensory-influenced choices. This pattern may involve the Lateral Intraparietal Cortex (LIP), given its proposed role in assessing and accumulating sensory information during perceptual decision-making¹⁵. While performance at high coherence remained stable, possibly due to ceiling effects, the significant improvement in choice accuracy observed specifically during low coherence trials may suggest greater reliance on prefrontal systems, including the dorsolateral Prefrontal Cortex (dlPFC), when sensory evidence is insufficient and reward history becomes more useful¹⁶. Because of the limited materials required to administer our test, our experiment could potentially be adapted for any brain-recording method such as fMRI, EEG, fNIRS, etc.

We would also expect to see involvement of both the Middle Temporal Visual Area (MT) and LIP, regions associated with motion perception and accumulation of sensory evidence, during trials with high coherence^{6, 4, 15}. The expected value associated with dot-motion direction may involve the ventral striatum due to its role in reinforcement learning¹⁷. Due to its role in working memory, the dlPFC may also show greater involvement during low coherence trials as participants attempt to use reward history and dot-motion direction¹⁶. The dlPFC could be analyzed in relation to the improved choice accuracy in low coherence trials from early (49.9%) to late (57.1%) trials (see Fig. 2). Strategy switching may also involve the Anterior Cingulate Cortex (ACC), particularly when subsequent trials switch from high to low or low to high coherence¹⁸. Given its proposed role in monitoring uncertainty, the ACC may also be relevant to the response-time difference between high coherence (M = 1804.57 ms) and low coherence (M = 1956.23 ms) trials (see Fig. 1)^{18, 5}.

Contributor Block

Programming: Coleman Klapheke (primary), Maxwell Rautmann (Code Verification)

Presentation: Claire Min (Slide Design, Intro), Maxwell Rautmann (Hypothesis, Concluding Remarks), Coleman Klapheke (Results, Graphs)

Introduction: Claire Min (Primary)

Methods: Coleman Klapheke (Statistical Analyses), Claire Min (Experimental Design)

Results: Coleman Klapheke (Primary), Claire Min (Figure Captions)

Discussion: Maxwell Rautmann (Primary), Coleman Klapheke (Summary of Results)

References: All members

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Supplementary Materials

Metric	Value
Number of Participants	[56]
Proportion of Choices Matching Learned Location Preference (Low Coherence)	[0.5994368178651065]
Proportion of Choices Matching Learned Location Preference (High Coherence):	[0.5773196487897485]
Proportion of Choices Matching Stimulus-Indicated Location (Low Coherence):	[0.5428571428571428]
Proportion of Choices Matching Stimulus-Indicated Location (High Coherence):	[0.5814285714285714]
Weight of Learned Location Preference on Choice (Low Coherence)	[0.35535714285714287]
Weight of Learned Location Preference on Choice (High Coherence)	[-0.06785714285714282]
Weight of Stimulus-Indicated Location on Choice (Low Coherence)	[0.8250000000000001]
Weight of Stimulus-Indicated Location on Choice (High Coherence)	[0.9928571428571429]
p-value for Change in Reliance on Learned vs Stimulus-Indicated Location Across Coherence	[0.0041466408924026865]
p-value for All Gender Differences in Decision Strategy	[nan]
p-value for Male/Female Gender Differences in Decision Strategy	[0.17926888303047386]
p-value for Age Group Differences in Decision Strategy	[0.5054716882054373]
Average Response Time Across Low Coherence	[1956.228376556158]
Average Response Time Across High Coherence	[1804.5651802834839]
p-value for Response Time Difference Across Coherence	[0.021264460035393576]

Table S1. This table summarizes the main behavioral and model-derived output variables, including participant count, low and high coherence choice alignment measures, model weights for learned box-location and stimulus-indicated reward location, response-time averages, and p-values from the primary decision-strategy, demographic, and response-time comparisons. Low-versus-high response-time differences were tested using paired-samples t-tests. Male/female decision-strategy differences were tested using independent-samples t-tests. Age-group decision-strategy differences were tested using one-way ANOVAs. The p-value for change in reliance on learned versus stimulus-indicated location across coherence reflects the coherence-dependent strategy-shift analysis comparing relative reliance on reward-history and stimulus-indicated choice information across coherence levels.

Metric	Mean $\pm$ SEM	Test Statistic	p-value	Effect Size
Number of Participants	56	—	—	—
Demographic Check: Overall Stimulus-Indicated Accuracy	M = 0.562 $\pm$ 0.022	M/F t = 0.604; Age F = 0.604	M/F p = 0.549; Age p = 0.615	—
Demographic Check: Overall Response Time	M = 1869.734 $\pm$ 148.965	M/F t = -0.638; Age F = 1.456	M/F p = 0.527; Age p = 0.237	—
Demographic Check: Strategy Shift	M = 0.061 $\pm$ 0.020	M/F t = 1.367; Age F = 0.793	M/F p = 0.179; Age p = 0.505	—
Bandit Validation: Overall Optimal Choices Above Chance	M = 0.562 $\pm$ 0.022	t(55) = 2.815	0.007	d = 0.376
Bandit Validation: Reward-History Alignment Above Chance	M = 0.588 $\pm$ 0.010	t(42) = 8.896	< .001	d = 1.357
Reward-History Choice Alignment: Low vs High Coherence	Low = 0.599 $\pm$ 0.013; High = 0.577 $\pm$ 0.012	t(42) = 1.446	0.156	d = 0.220
Stimulus-Indicated Choice Alignment: Low vs High Coherence	Low = 0.547 $\pm$ 0.024; High = 0.587 $\pm$ 0.028	t(42) = -2.155	0.037	d = -0.329
Reward-History vs Stimulus Alignment: Low Coherence	Reward History = 0.599 $\pm$ 0.013; Stimulus = 0.547 $\pm$ 0.024	t(42) = 1.853	0.071	d = 0.283
Reward-History vs Stimulus Alignment: High Coherence	Reward History = 0.577 $\pm$ 0.012; Stimulus = 0.587 $\pm$ 0.028	t(42) = -0.299	0.766	d = -0.046
Learned Box-Location Regression Weight: Low vs High Coherence	Low = 0.355 $\pm$ 0.450; High = -0.068 $\pm$ 0.439	t(55) = 1.092	0.280	d = 0.146
Stimulus Regression Weight: Low vs High Coherence	Low = 0.825 $\pm$ 0.433; High = 0.993 $\pm$ 0.327	t(55) = -0.493	0.624	d = -0.066
Box-Location vs Stimulus Regression Weight: Low Coherence	Box = 0.355 $\pm$ 0.450; Stimulus = 0.825 $\pm$ 0.433	t(55) = -0.761	0.450	d = -0.102
Box-Location vs Stimulus Regression Weight: High Coherence	Box = -0.068 $\pm$ 0.439; Stimulus = 0.993 $\pm$ 0.327	t(55) = -1.992	0.051	d = -0.266
Response Time: Low vs High Coherence	Low = 1956.228 $\pm$ 165.691; High = 1804.565 $\pm$ 141.572	t(55) = 2.371	0.021	d = 0.317
Learning Across Task: Low Coherence Early vs Late	Early = 0.499 $\pm$ 0.022; Late = 0.571 $\pm$ 0.027	t(55) = -3.106	0.003	d = -0.415
Learning Across Task: High Coherence Early vs Late	Early = 0.573 $\pm$ 0.029; Late = 0.604 $\pm$ 0.030	t(55) = -1.041	0.302	d = -0.139
Four-Level Coherence Trend: Stimulus minus Box Weight	Slope = 4.645 $\pm$ 1.716	t(55) = 2.707	0.009	d = 0.362
Gender Identity Counts	Male: 33, Female: 22, Non-binary: 1, Other: 0	—	—	—
Age Group Counts	21 and younger: 12, 22-30: 32, 31-50: 8, 51 and older: 4	—	—	—

Table S2. This table displays participant-level means $\pm$ SEM, test statistics, p-values, and effect sizes for the main analyses used in the Results section. One-sample t-tests were used for chance-level validation of overall optimal choices and reward-history alignment against 0.50. Paired-samples t-tests were used for low-versus-high coherence comparisons, reward-history versus stimulus-indicated alignment comparisons, response-time comparisons, early-versus-late learning comparisons, and model-weight comparisons. The four-level coherence trend was tested using one-sample t-tests on participant-level slopes for stimulus-minus-box model weights across the four coherence levels. Demographic checks used independent-samples t-tests for male/female comparisons and one-way ANOVAs for age-group comparisons.